

Advanced Chooser Option Pricing Model

Final Project Report

JPM underlying | Week 1 through Week 8

This report documents the complete data, pricing, validation, machine-learning, sensitivity, and application pipeline. The key result is a working dual-pricing tool whose selected volatility model improves test proxy RMSE while retaining BSM as the pricing rule.

Executive Summary

The project developed a Black-Scholes-Merton simple chooser pricing model for JPM and extended it with two machine-learning approaches. Weeks 1 and 2 collected, aligned, cleaned, and engineered JPM, VIX, and Treasury data. Week 3 replicated the paper-matched chooser model with $K=\$150$, $T1=0.5$, and $T2=1.0$. Week 4 established the baseline. Weeks 5 and 6 built and compared ML models. Week 7 added stress tests and live-data infrastructure. Week 8 packaged the full chain into a Streamlit application.

Final pricing route	MAE	RMSE	R ²
Week 4 BSM baseline	3.5698	4.8706	0.9460
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575
Approach 2 direct price	2.8501	4.5404	0.9530

Approach 1 reduces historical test proxy RMSE by 11.30% versus BSM. The repaired bounded direct route has RMSE 4.5404 and MAE 2.8501. The revision reuses the historical test retrospectively after a live extrapolation failure. Targets remain ex-post forward-volatility BSM proxies, not observed chooser trades or JPM stock prices.

Submission Status

- Technical Weeks 1–8 complete
- Final application and model bundle complete
- Narrated demo recording remains pending; public hosting is optional

1 Project Objective and Contract

A simple chooser option allows the holder to decide at time T_1 whether the contract becomes a European call or a European put with final maturity T_2 . The project uses JPM as the underlying and retains the paper-matched contract so results remain comparable across weeks.

Parameter	Value	Treatment
Strike K	\$150	Required project setting
Choice time T_1	0.5 year	Six-month decision
Maturity T_2	1 year	Required setting
Dividend yield q	2.33%	Paper value retained
Spot, rate, volatility	Date dependent	Week 2 data or current snapshot

Project Chain

Week 1 raw market data -> Week 2 preprocessing and features -> Week 3 chooser BSM -> Week 4 baseline evaluation -> Week 5 ML design -> Week 6 training and interpretation -> Week 7 stress tests and data updater -> Week 8 final tool.

Each stage consumes the previous stage's files. This design prevents the later work from replacing the original market-data contribution or changing the Week 3 pricing economics.

2 Market Data and Feature Engineering

The Week 2 dataset contains 1,760 aligned trading-date observations from 2018-01-02 through 2024-12-30 and 23 cleaned or engineered columns. JPM OHLCV data supplies spot and return information. VIX represents market volatility conditions. Treasury yields provide the risk-free-rate proxy.

Feature group	Examples	Use
Returns and volatility	Daily return, log return, rolling vol	BSM input and ML prediction
Trend and momentum	MA20, MA50, price momentum	Market state
Risk indicators	VIX, VIX return, JPM-VIX correlation	Regime and ML features
Rates	Treasury rate, rate momentum	Discounting and ML features
Trading activity	Intraday range, volume z-score	Short-horizon conditions

Feature construction preserves dates and units. Treasury percentages are divided by 100 before BSM use. Rolling volatility is annualized with 252 trading days. Features use information available on or before each row date. Because each target uses the following 20 trading days, the final 20 rows before both development boundaries are excluded from fitting and selection.

GitHub Actions now reproduces the Week 2 pipeline from the archived Week 1 inputs every Monday and on manual request. The verified run rebuilt the processed data, checked its schema and date range, and uploaded the outputs as an artifact. The project repository is <https://github.com/Frank-Wang-Jc/jpm-chooser-option-pricing>.

3 BSM Chooser Model

Week 3 implemented European call and put functions, the adjusted choice boundary, and the simple chooser decomposition. The analytical price equals a European call expiring at T2 plus an adjusted put component whose strike reflects the remaining period after T1.

Validation test	Result
Put-call parity error	0.000000
Paper choice decisions	10 of 10 matched
Paper payoff calculations	Maximum error approximately 0
Monte Carlo vs closed form	0.823 standard errors
Week 3 vs Week 4 regression	Maximum difference 0.000000

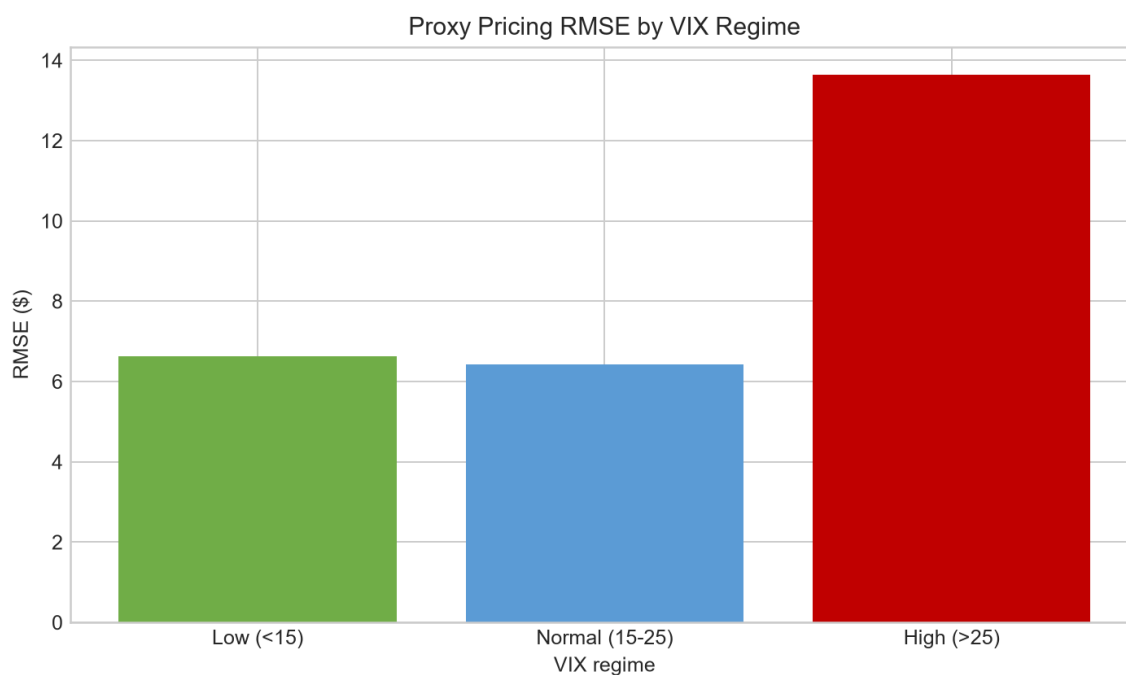
Under the paper inputs, the analytical chooser price was \$29.1299. Under the latest Week 2 baseline, the value was approximately \$82.53 because JPM was substantially above the fixed strike. The higher level is a contract-moneyness result, not a change to the formula.

4 Baseline Evaluation

Week 4 separated computational evidence from market evidence. The historical proxy compares trailing-volatility pricing with a chooser price computed from the next 20 days of realized volatility. The listed market check applies vanilla BSM to delayed JPM call and put quotes.

Benchmark	MAE	RMSE	R ²
Historical chooser proxy	3.7757	8.3567	0.8242
Listed vanilla, Week 2 vol	0.3635	0.4401	0.9845
Listed vanilla, quoted IV	0.1658	0.2201	0.9961

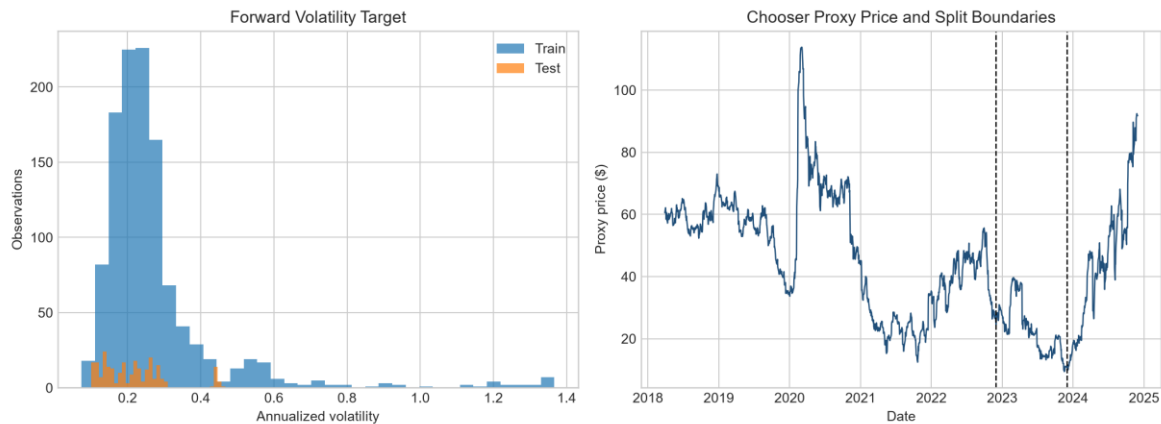
Figure 1 Historical proxy RMSE by VIX regime



5 Machine Learning Design

Approach 1 predicts forward volatility from 17 features and passes it through BSM. Approach 2 retains 19 observable inputs but learns normalized time value within discounted spot-and-strike bounds. Constrained moneyness tails replace competing dollar and log-price coefficients. Inference calls neither BSM nor a volatility forecaster.

Figure 2 Target construction and chronological split

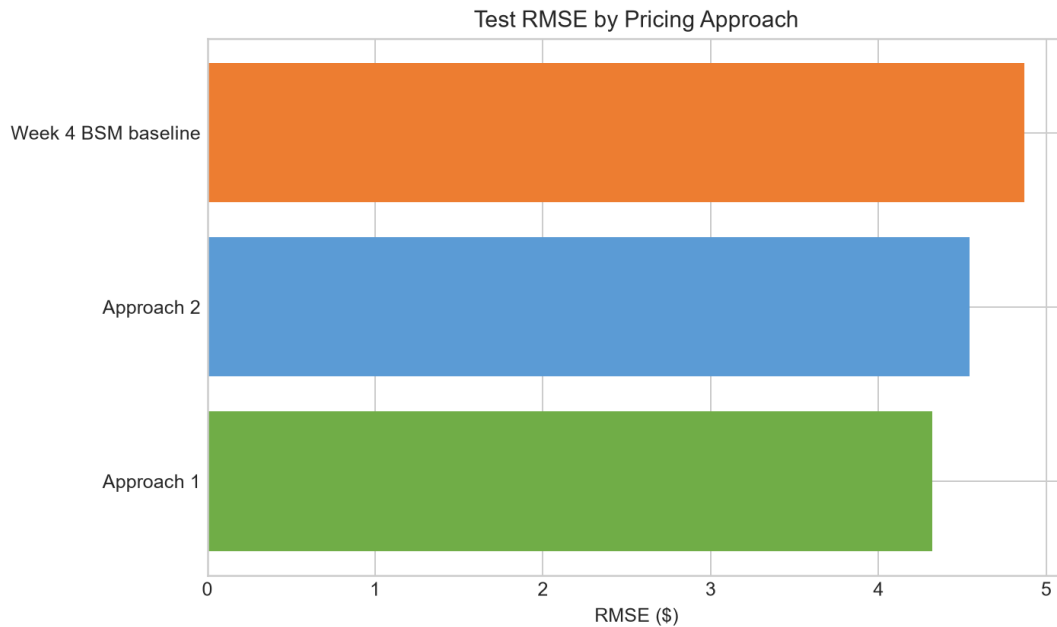


Use	Rows	Period
Train	1,156	2018-03-29 to 2022-10-28
Validation	232	2022-11-29 to 2023-10-31
Purged	40	20 rows before each boundary
Test	252	2023-11-30 to 2024-11-29

6 Training and Test Performance

The original candidate comparison selected Random Forest for Approach 1. After a negative direct-price live prediction, purged training CV tuned bounded time-value regression. The revised historical test is a retrospective comparison, not a fresh holdout. RF plus BSM still has the lowest test RMSE; repaired direct pricing has the lowest MAE.

Figure 3 Historical test proxy RMSE



Volatility forecast	MAE	RMSE
Persistence	0.085584	0.120650
Selected Random Forest	0.072424	0.094069

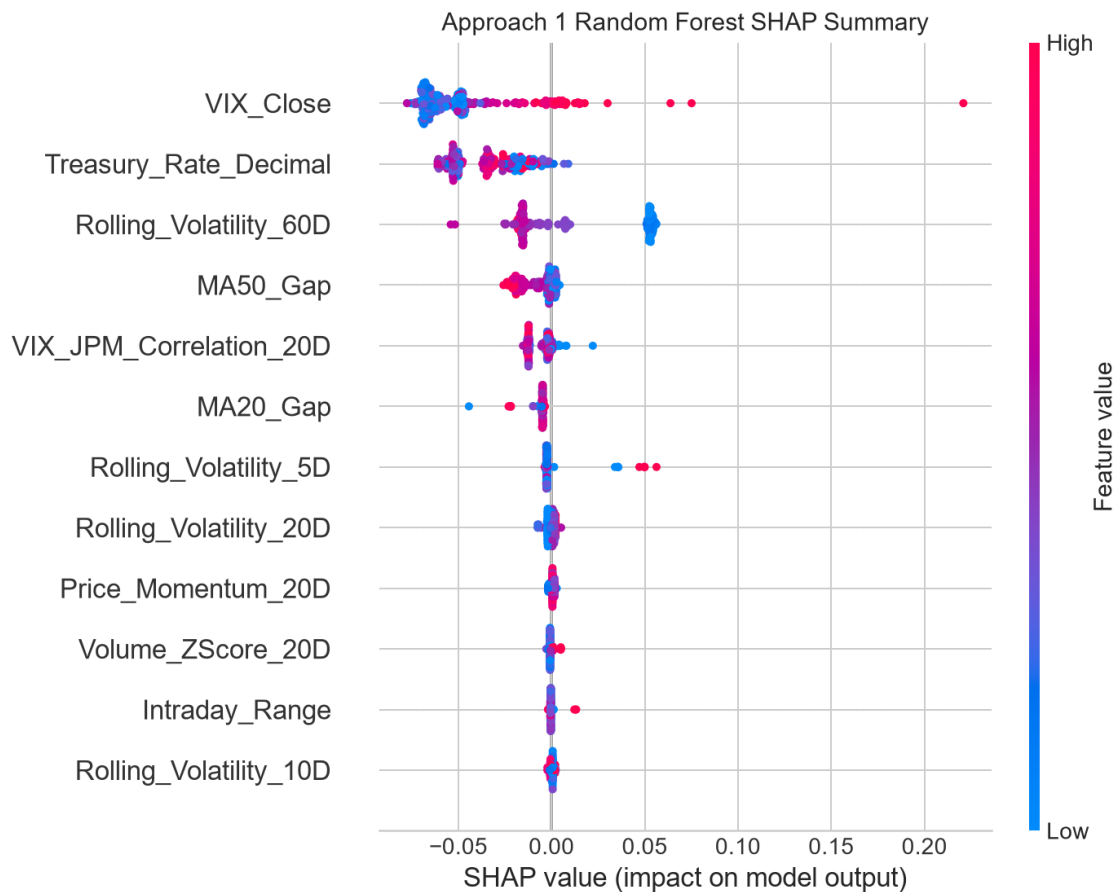
The legacy linear model produced -\$111.56 at the September 10 snapshot, concealed by zero clipping. Its replacement enforces structural bounds without clipping output. Test RMSE improves from 49.9891 to 4.5404, while validation RMSE worsens from 5.4837 to 7.3983. A post-freeze 404-row 2025–2026 proxy diagnostic gives RMSE 3.7972 using later downloaded adjusted history. It is not point-in-time market-price validation.

Both selected Week 6 estimators are archived as native Python pickle files. Each file was reloaded and checked on all 252 historical test rows; the maximum absolute difference from the saved Week 6 predictions was below 1e-12. Joblib runtime copies and model metadata are bundled with the Week 8 application.

7 Model Interpretation

Tree SHAP explains the original Random Forest volatility predictor. Training-background permutation SHAP explains the complete bounded direct-price function in dollar units. The direct model excludes current BSM price and forward-volatility targets from its input features.

Figure 4 Approach 1 SHAP feature importance



VIX-related mean absolute SHAP shares are 40.80% for Approach 1 and 23.29% for repaired Approach 2. Compare relative shares within each model because the explanation targets and units differ. These sample-based attributions are not causal effects.

8 Sensitivity and Stress Testing

Week 7 implemented the required 50% volatility spike and absolute 2 percentage-point rate hike. The structural BSM test used the last full Week 5 model row, which had JPM spot \$241.12 and historical volatility 45.72%.

Figure 5 Required Week 7 stress scenarios



Scenario	BSM price	Change
Base	\$99.2617	\$0.0000
50% volatility spike	\$115.6683	+\$16.4066
2% rate hike	\$101.3317	+\$2.0700
Combined	\$116.9177	+\$17.6561

9 Data Update and Final Application

The updater rebuilds all original ML features from JPM OHLCV, Cboe VIX, and FRED DGS10 histories at a common completed date. DGS10 preserves the trained rate feature; DGS1 supplies the one-year pricing rate. Requests have bounded retries and independent source caches. The application refreshes on opening, every 15 minutes while active, and on manual request, retaining a labeled complete snapshot when required data are unavailable.

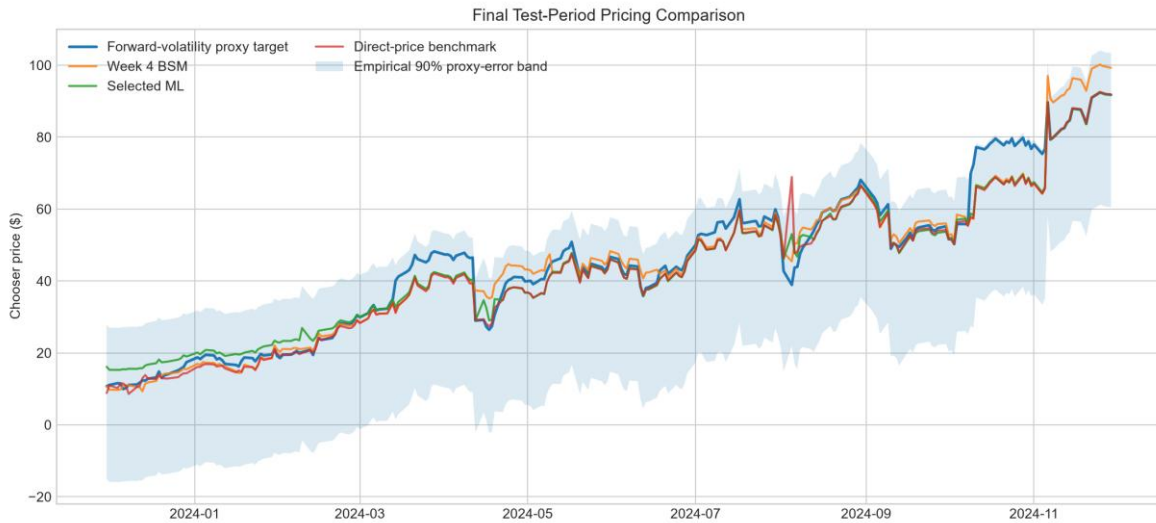
Application tab	Content
Dual pricing	BSM, selected ML-volatility route, direct-price benchmark, Greeks, proxy error band
Sensitivity	Volatility curve, rate curve, required stress scenarios
Model performance	MAE, RMSE, R^2 and historical test price trend
Methods and limits	Target definition, feature date, model boundary and interpretation

The application restricts ML pricing to $K=\$150$, $q=2.33\%$, $T1=0.5$, and $T2=1.0$ because those values define the training target. BSM remains available for other valid contracts. The automated Streamlit test rendered four tabs, seven metrics, and one refresh control without exceptions.

10 Error Margin and Live Extrapolation

The selected route retains its 5th and 95th percentile residual band from 924 purged expanding-window training predictions. The later 252-row test has 99.60% coverage and mean width \$42.8744. This existing historical evaluation is retained during the model revision; no new untouched test claim is made.

Figure 6 Final test-period pricing and proxy-error band



The reproducible offline example uses the complete 2024-12-30 feature snapshot. Its \$231.08 spot exceeds the fitted maximum \$152.49, so the warning remains. The repaired direct model enforces pointwise bounds, which do not guarantee an arbitrage-free surface or out-of-range market accuracy. Live refresh updates features without retraining.

11 Limitations and Controls

Limitation	Control
No public chooser transaction history	Label the target and interval as theoretical proxies
Treasury maturity mismatch in Week 2	Document the proxy and add DGS1 to live updates
Dividend yield not collected	Retain the paper's 2.33% assumption
Tree-model extrapolation	Keep BSM visible and flag values outside training ranges
Different source publication dates	Rebuild every feature at a common completed date and label cached-source use
No reproducible sentiment series	Record the data gap; do not relabel VIX

These controls keep the final interface honest about what the evidence supports. The tool is a research prototype and does not provide trading or investment advice.

Data and Software Sources

- Project brief and supplied Week 1–3 files
- Week 2 processed JPM, VIX, and Treasury dataset
- Cboe delayed listed JPM option quotes and VIX history
- FRED DGS10 and DGS1 series
- Python, NumPy, pandas, SciPy, scikit-learn, XGBoost, SHAP, and Streamlit

12 Conclusion and Submission Checklist

The project now contains a continuous and reproducible Week 1–8 pipeline. The Week 3 chooser model was preserved, Week 4 established a defensible baseline, and the selected Week 6 volatility model improved test proxy RMSE. Week 7 verified the required shocks and the data updater. Week 8 packaged the pricing routes, out-of-fold calibrated error band, performance evidence, and limitations in a tested application.

Submission item	Status
Code, executed notebooks, GitHub repository, and Actions	Complete; repository integrity, Streamlit smoke tests, and scheduled preprocessing passed
Parameter and model files	Complete
Validation, benchmark, sensitivity, and weekly reports	Complete
Final 10–15 page report	Complete
Final presentation	Complete
Deployable pricing tool and README	Complete; local execution verified
Narrated 5–10 minute MP4	Student recording action

Recommended Demonstration Flow

Open the app, confirm the source dates, explain the default contract, compare BSM and ML prices, show the proxy-error interval, run the required stress scenarios, review the test metrics, and finish with the target and extrapolation limitations. The Week 8 folder contains a timed script and recording checklist.